

Magnetism

MCQ

1. Magnetism is a force of

- A) Friction
- B) Attraction or repulsion
- C) Gravity
- D) Pressure

Answer: B

2. A magnet can attract certain types of

- A) Wood
- B) Plastic
- C) Metals
- D) Glass

Answer: C

3. Magnetism does not require

- A) Contact
- B) Motion
- C) Energy
- D) Direction

Answer: A

4. Magnetism is caused by the motion of

- A) Protons
- B) Neutrons
- C) Free electrons
- D) Nuclei

Answer: C

5. A magnet is an object that can

- A) Only repel
- B) Only attract
- C) Attract and repel

D) Destroy metals

Answer: C

6. Unlike poles of magnets

A) Repel each other

B) Attract each other

C) Remain neutral

D) Disappear

Answer: B

7. Like poles of magnets

A) Attract each other

B) Do not interact

C) Repel each other

D) Melt

Answer: C

8. Magnetism is related to

A) Heat energy

B) Electric charges

C) Sound waves

D) Light rays

Answer: B

9. Which of the following uses magnets?

A) Cupboard doors

B) Paper clips

C) Wooden table

D) Plastic chair

Answer: A

10. Debit and credit cards use magnets for

A) Decoration

B) Identification

C) Data storage

D) Power supply

Answer: C

11. Magnets used in hospitals are found in

- A) ECG
- B) X-ray
- C) MRI
- D) CT scan

Answer: C

12. Magnets found naturally are called

- A) Artificial magnets
- B) Temporary magnets
- C) Natural magnets
- D) Induced magnets

Answer: C

13. Natural magnets are always

- A) Temporary
- B) Weak
- C) Permanent
- D) Artificial

Answer: C

14. Natural magnets are usually found in

- A) Rocks only
- B) Sandy deposits
- C) Oceans
- D) Forests

Answer: B

15. Lodestone is also known as

- A) Hematite
- B) Siderite
- C) Magnetite
- D) Ferrite

Answer: C

16. Chemical formula of magnetite is

- A) FeO

B) Fe_2O_3

C) Fe_3O_4

D) FeSO_4

Answer: C

17. Magnetite is an ore of

A) Copper

B) Aluminium

C) Iron

D) Zinc

Answer: C

18. Which iron ore contains the highest percentage of iron?

A) Hematite

B) Siderite

C) Magnetite

D) Ferrite

Answer: C

19. Artificial magnets are made by

A) Nature

B) Magnetising materials

C) Cooling metals

D) Heating plastics

Answer: B

20. Artificial magnets are also called

A) Natural magnets

B) Induced magnets

C) Man-made magnets

D) Weak magnets

Answer: C

21. Which is an example of an artificial magnet?

A) Lodestone

B) Magnetite

C) Bar magnet

D) Hematite

Answer: C

22. Artificial magnets are stronger than

A) Steel

B) Temporary magnets

C) Natural magnets

D) Iron

Answer: C

23. Which metal is commonly used to make magnets?

A) Aluminium

B) Iron

C) Copper

D) Lead

Answer: B

24. Neodymium magnets are known for being

A) Weak

B) Temporary

C) Strongest

D) Natural

Answer: C

25. Permanent magnets retain magnetism for

A) Short time

B) Long time

C) No time

D) Only in current

Answer: B

26. ALNICO is an alloy of

A) Aluminium, nickel and cobalt

B) Aluminium, iron and copper

C) Nickel, zinc and iron

D) Iron, carbon and chromium

Answer: A

27. Temporary magnets lose magnetism when

- A) Heated
- B) Current is removed
- C) Painted
- D) Polished

Answer: B

28. Electromagnets are examples of

- A) Natural magnets
- B) Permanent magnets
- C) Temporary magnets
- D) Weak magnets

Answer: C

29. Electric bells use

- A) Permanent magnets
- B) Natural magnets
- C) Temporary magnets
- D) No magnets

Answer: C

30. Magnetic cranes use

- A) Bar magnets
- B) Temporary magnets
- C) Natural magnets
- D) Ring magnets

Answer: B

31. Every magnet has

- A) One pole
- B) Three poles
- C) Two poles
- D) No poles

Answer: C

32. The ends of a magnet where attraction is strongest are called

- A) Axes

B) Centres

C) Poles

D) Fields

Answer: C

33. Magnetic poles always exist in

A) Single form

B) Pairs

C) Groups

D) Circles

Answer: B

34. Splitting a magnet vertically results in

A) No magnetism

B) Single pole

C) Two magnets

D) Loss of poles

Answer: C

35. Directive property of a magnet means it

A) Attracts iron

B) Repels magnets

C) Aligns North–South

D) Loses magnetism

Answer: C

36. A freely suspended magnet points in

A) East–West direction

B) North–South direction

C) Any direction

D) Circular path

Answer: B

37. Magnets can lose their properties due to

A) Proper storage

B) Heating

C) Polishing

D) Painting

Answer: B

38. Dropping a magnet repeatedly causes

A) Increase in strength

B) No effect

C) Loss of magnetism

D) Change of colour

Answer: C

39. Magnetic field is the region around a magnet where

A) Light bends

B) Heat spreads

C) Magnetic force acts

D) Sound travels

Answer: C

40. Unit of magnetic field is

A) Watt

B) Tesla

C) Joule

D) Newton

Answer: B

41. One tesla is equal to

A) 100 gauss

B) 1,000 gauss

C) 10,000 gauss

D) 100,000 gauss

Answer: C

42. A magnetic compass contains

A) A glass needle

B) A steel rod

C) A pivoted magnet

D) An electromagnet

Answer: C

43. Compass needle points towards

- A) Geographic east
- B) Geographic west
- C) Geographic north
- D) Geographic south

Answer: C

44. Magnetic field lines are

- A) Straight lines
- B) Curved lines
- C) Broken lines
- D) Zigzag lines

Answer: B

45. Magnetic field lines emerge from

- A) South pole
- B) Both poles
- C) North pole
- D) Centre

Answer: C

46. Magnetic field lines enter the magnet at

- A) North pole
- B) South pole
- C) Centre
- D) Axis

Answer: B

47. Materials attracted by magnets are called

- A) Non-magnetic
- B) Insulators
- C) Magnetic materials
- D) Conductors

Answer: C

48. Which is a magnetic material?

- A) Copper

B) Aluminium

C) Iron

D) Brass

Answer: C

49. Which is a non-magnetic material?

A) Nickel

B) Cobalt

C) Steel

D) Copper

Answer: D

50. Magnetically soft materials are

A) Hard to magnetise

B) Easily magnetised

C) Non-magnetic

D) Always permanent

Answer: B

51. Diamagnetic materials are

A) Strongly attracted

B) Weakly attracted

C) Not attracted

D) Permanently magnetised

Answer: C

52. Diamagnetic materials align

A) Parallel to field

B) Perpendicular to field

C) Circularly

D) Randomly

Answer: B

53. Paramagnetic materials are

A) Strongly repelled

B) Weakly magnetised

C) Permanently magnetised

D) Non-magnetic

Answer: B

54. Paramagnetic materials lose magnetism when

A) Heated

B) Cooled

C) Field is removed

D) Hammered

Answer: C

55. Ferromagnetic materials are

A) Weakly attracted

B) Not attracted

C) Strongly attracted

D) Repelled

Answer: C

56. Which is a ferromagnetic material?

A) Aluminium

B) Copper

C) Iron

D) Brass

Answer: C

57. Ferromagnetic materials can be

A) Never magnetised

B) Permanently magnetised

C) Weakly magnetised

D) Non-magnetic

Answer: B

58. On heating, ferromagnetic materials become

A) Diamagnetic

B) Paramagnetic

C) Non-magnetic

D) Superconducting

Answer: B

59. Curie temperature is related to

- A) Diamagnetism
- B) Paramagnetism
- C) Ferromagnetism
- D) Electricity

Answer: C

60. Earth behaves like a

- A) Electric dipole
- B) Magnetic dipole
- C) Bar magnet only
- D) Ring magnet

Answer: B

61. Earth's magnetic south pole lies near

- A) Geographic south
- B) Geographic east
- C) Geographic north
- D) Geographic west

Answer: C

62. Magnetic axis of Earth is inclined by about

- A) 0°
- B) 5°
- C) 10° – 15°
- D) 45°

Answer: C

63. Earth's magnetism is partly due to

- A) Water
- B) Molten metals
- C) Atmosphere
- D) Clouds

Answer: B

64. The inner core of the Earth is

- A) Liquid

B) Gaseous

C) Solid

D) Hollow

Answer: C

65. The outer core of the Earth is

A) Solid

B) Liquid

C) Gaseous

D) Empty

Answer: B

66. Magnetic field strength of Earth ranges in

A) Tesla

B) Millitesla

C) Microtesla

D) Kilotesla

Answer: C

67. Compasses work due to

A) Earth's gravity

B) Earth's magnetism

C) Wind direction

D) Sunlight

Answer: B

68. In ancient times magnets were used to

A) Store data

B) Generate power

C) Find directions

D) Cut metals

Answer: C

69. Dynamos use magnets to

A) Store energy

B) Generate electricity

C) Reduce heat

D) Increase speed

Answer: B

70. Loudspeakers use magnets to convert

A) Sound to light

B) Heat to sound

C) Electrical signals to sound

D) Motion to light

Answer: C

71. Maglev trains float using

A) Gravity

B) Friction

C) Powerful electromagnets

D) Permanent magnets only

Answer: C

72. Magnetic conveyor belts are used to

A) Lift people

B) Sort materials

C) Cool machines

D) Heat metals

Answer: B

73. Hard disks store data using

A) Electric charges

B) Light rays

C) Magnetic patterns

D) Heat energy

Answer: C

74. MICR numbers on cheques are read using

A) Light

B) Sound

C) Magnets

D) Heat

Answer: C

75. Screwdriver tips are magnetised to

- A) Cut screws
- B) Attract hands
- C) Hold screws
- D) Heat screws

Answer: C

76. MRI uses

- A) Natural magnets
- B) Weak magnets
- C) Electromagnets
- D) Temporary iron rods

Answer: C

77. Magnetic field lines never

- A) Cross each other
- B) Curve
- C) Form loops
- D) Have direction

Answer: A

78. Strength of magnetic field is maximum near

- A) Centre
- B) Axis
- C) Poles
- D) Outside region

Answer: C

79. Magnetic materials contain

- A) No electrons
- B) Aligned domains
- C) Neutral atoms only
- D) No charges

Answer: B

80. Magnetism is an example of a

- A) Contact force
- B) Non-contact force
- C) Mechanical force
- D) Frictional force

Answer: B